

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 24]

**If two functions  $f(x)$  and  $g(x)$  are differentiable, then their product is differentiable and the two functions satisfy the product rule:  $(f(x)g(x))' = f'(x)g(x) + g(x)f'(x)$ .**

**In this question, you will meet examples of pairs of differentiable functions,  $f(x)$  and  $g(x)$ , that also satisfy  $(f(x)g(x))' = f'(x)g'(x)$ .**

In part (a), consider  $f(x) = \frac{1}{(2-x)^2}$ , where  $x \in \mathbb{R}$ ,  $x \neq 2$ , and  $g(x) = x^2$ , where  $x \in \mathbb{R}$ .

(a) (i) Find an expression for  $f'(x)$ . [2]

(ii) Show that  $f'(x)g'(x) = \frac{4x}{(2-x)^3}$ . [2]

(iii) Show that  $f(x)g'(x) + g(x)f'(x) = \frac{4x}{(2-x)^3}$ . [4]

In parts (b) and (c), consider two non-constant functions,  $f(x)$  and  $g(x)$ , where  $f(x) > 0$  and  $g(x) \neq g'(x)$ .

(b) By rearranging the equation  $f(x)g'(x) + g(x)f'(x) = f'(x)g'(x)$ , show that  $\frac{f'(x)}{f(x)} = \frac{g'(x)}{g'(x) - g(x)}$ . [2]

(This question continues on the following page)

**(Question 1 continued)**

- (c) Hence, by integrating both sides of  $\frac{f'(x)}{f(x)} = \frac{g'(x)}{g'(x) - g(x)}$ , show that  $f(x) = Ae^{\left(\int \frac{g'(x)}{g'(x) - g(x)} dx\right)}$ , where  $A$  is an arbitrary positive constant. [2]

The result from part (c) can be used to find pairs of functions,  $f(x)$  and  $g(x)$ , which satisfy **both** of the following:

$$(f(x)g(x))' = f(x)g'(x) + g(x)f'(x) \text{ and } (f(x)g(x))' = f'(x)g'(x).$$

In parts (d) and (e), use the result in part (c) with  $A = 1$ .

- (d) Consider  $g(x) = xe^x$ .

Find  $f(x)$  such that  $f(x)$  and  $g(x)$  satisfy the above two equations. [5]

- (e) Consider  $g(x) = \sin x + \cos x$ .

Find  $f(x)$  such that  $f(x)$  and  $g(x)$  satisfy the above two equations over the domain  $0 < x < \pi$ .

Give your answer in the form  $f(x) = \sqrt{e^x h(x)}$ , where  $h(x)$  is a function to be determined. [7]